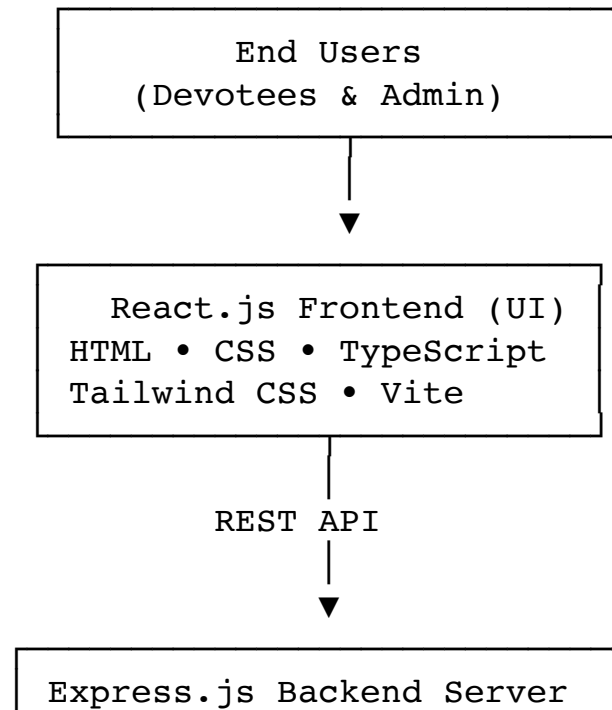


Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	15 July 2026
Team ID	XXXXX
Project Name	Darshanease
Maximum Marks	4 Marks

Technical Architecture



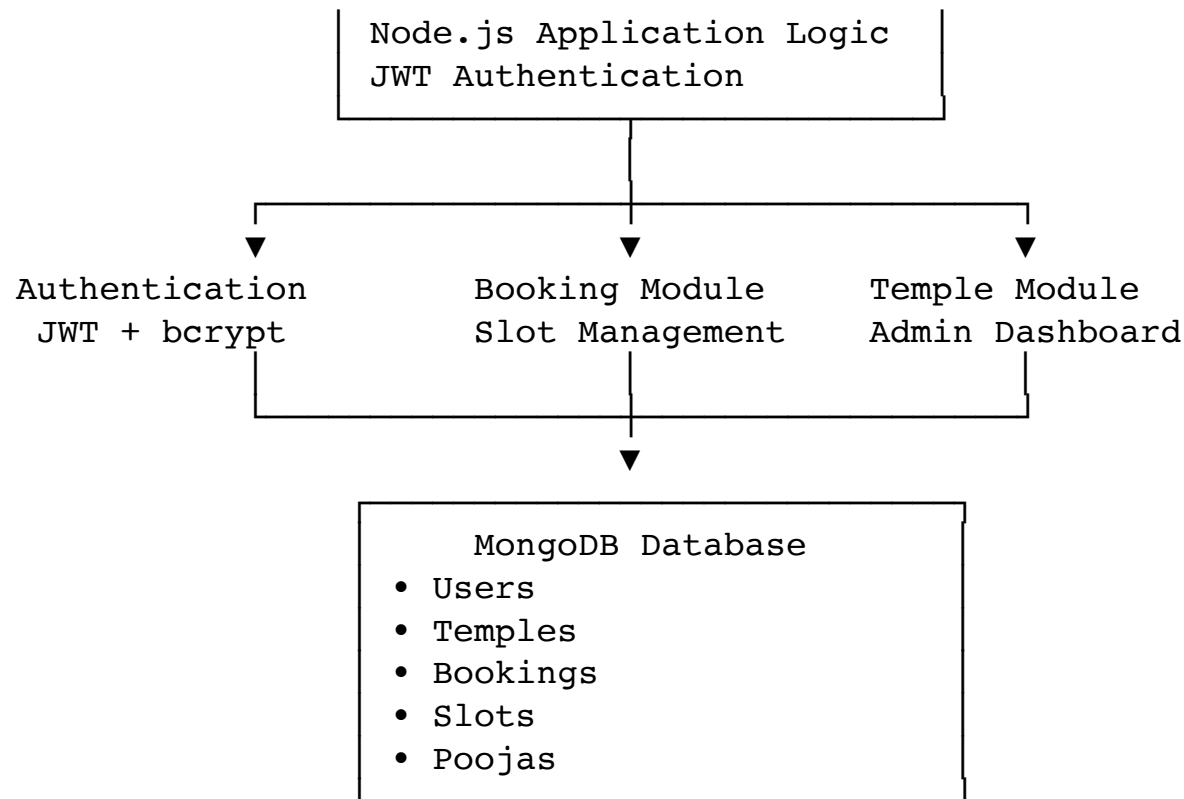


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive web interface for devotees and administrators	React.js, TypeScript, HTML5, CSS3, Tailwind CSS, Vite
2	Application Logic-1	User Registration and Authentication	Node.js, Express.js, JWT, bcrypt
3	Application Logic-2	Temple Management and Darshan Slot Booking	Express.js, RESTful APIs

4	Application Logic-3	Booking Management, Admin Dashboard, and Business Logic	Node.js, Express.js
5	Database	Stores users, temples, bookings, pooja details, and darshan slots	MongoDB (NoSQL), Mongoose
6	Cloud Database	Cloud-hosted database for production deployment	MongoDB Atlas
7	File Storage	Stores temple images and static assets	Local File System (Future: Cloudinary / AWS S3)
8	External API-1	Maps and navigation services (Future Enhancement)	Google Maps API
9	External API-2	Online payment processing (Future Enhancement)	Razorpay API / Stripe API
10	Machine Learning Model	Not implemented in the current version (Future: Crowd Prediction)	Machine Learning (Future Scope)
11	Infrastructure (Server / Cloud)	Local development and cloud deployment	Local Server, Render (Backend), Vercel (Frontend), MongoDB Atlas

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frontend and backend developed using open-source technologies	React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Vite
2	Security Implementations	Password encryption, JWT authentication, protected routes, environment variables	bcrypt, JWT, HTTPS, dotenv, Express Middleware
3	Scalable Architecture	Three-tier architecture with separate frontend, backend, and database layers	MERN Stack, RESTful APIs
4	Availability	Application can be deployed on cloud for 24×7 availability	Vercel, Render, MongoDB Atlas
5	Performance	Fast frontend rendering, optimized API communication, and efficient database queries	React Virtual DOM, Vite, Axios, MongoDB Indexing

Technology Stack Summary

Layer	Technology
Frontend	React.js, TypeScript, HTML5, CSS3, Tailwind CSS, Vite
Backend	Node.js, Express.js
Database	MongoDB, Mongoose
Authentication	JWT, bcrypt
API	RESTful APIs
State Management	React Context API
Package Manager	pnpm / npm
Version Control	Git, GitHub
IDE	Visual Studio Code
Deployment	Vercel (Frontend), Render (Backend), MongoDB Atlas

Architecture Highlights

- Uses a **Three-Tier Architecture** consisting of Presentation Layer (Frontend), Business Logic Layer (Backend), and Database Layer (MongoDB).
- Implements **RESTful APIs** for seamless communication between the React frontend and the Express.js backend.
- Secures user authentication using **JSON Web Token (JWT)** and encrypts passwords with **bcrypt**.

- Utilizes **MongoDB**, a NoSQL database, for efficient storage of users, temples, bookings, pooja details, and darshan slots.
- Provides a **responsive user interface** built with React.js and Tailwind CSS, ensuring compatibility across desktop, tablet, and mobile devices.
- Supports a **modular and scalable MERN architecture**, allowing easy integration of future features such as online payment gateways, QR code-based ticket verification, AI-based crowd prediction, Google Maps integration, and cloud deployment.
- Can be deployed using modern cloud platforms such as **Vercel**, **Render**, and **MongoDB Atlas**, providing high availability and scalability.

References

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